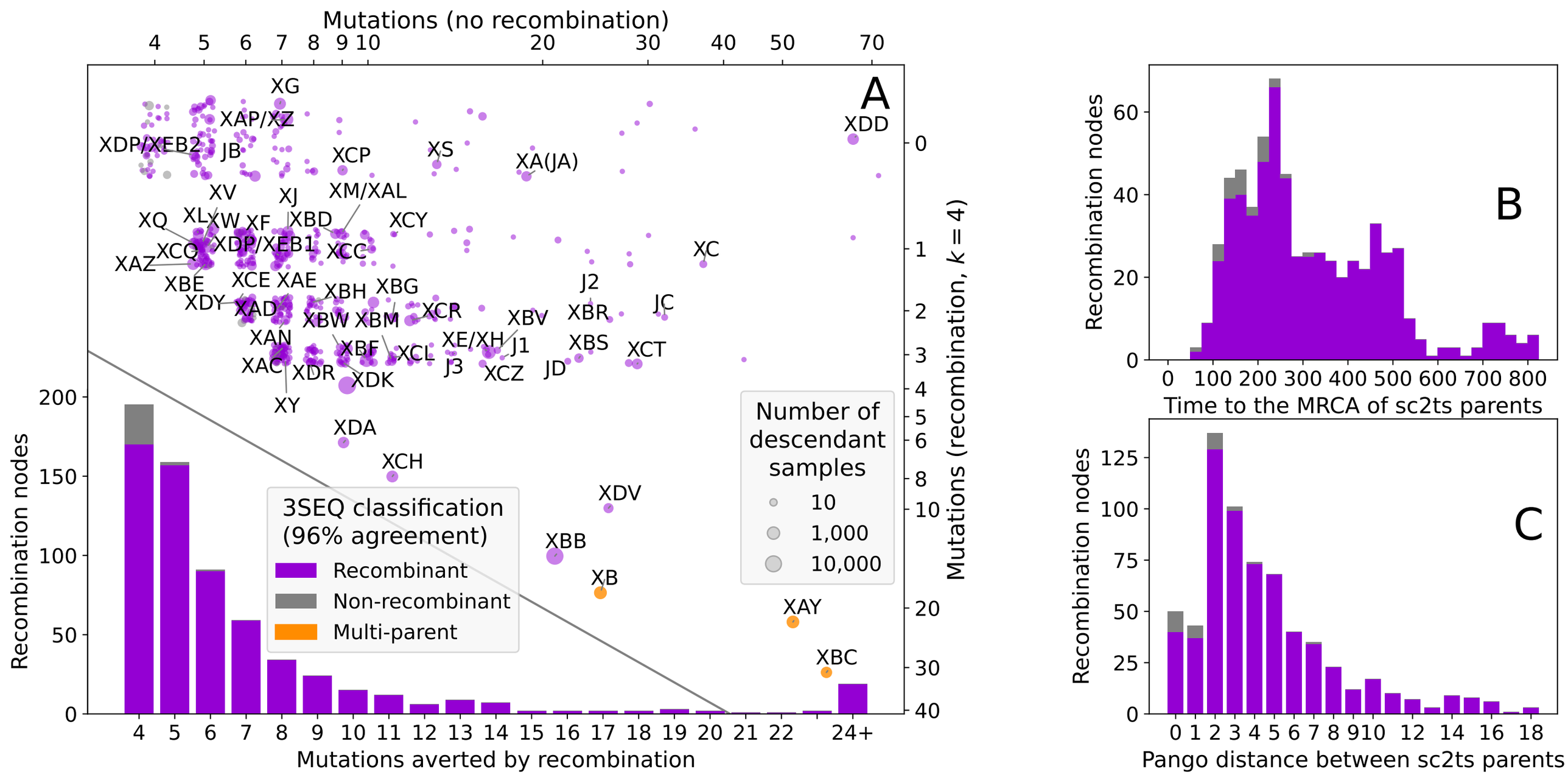

A
Mutations (no recombination)
4 5 6 7 8 9 10 20 30 40 50 70
Recombination nodes
Mutations averted by recombination
Mutations (recombination, k = 4)
4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 24+
0 50 100 150 200
0 1 2 3 4 5 6 8 10 20 30 40
XG
XAP/XZ
XDP/XEB2
JB
XCP
XS
XA(JA)
XDD
XV
XJ
XM/XAL
XQ
XL
XW
XF
XBD
XCY
XDP/XEB1
XCC
XC
XCQ
XAZ
XBE
XCE
XAE
XBH
XBG
J2
XDY
XAD
XCR
XBV
XBR
JC
XBW
XBM
XAN
XE/XH
J1
XBS
XCT
XBF
XCL
XAC
XDR
XDK
J3
XCZ
JD
XY
XDA
XCH
XDV
XBB
XB
XAY
XBC
3SEQ classification
(96% agreement)
Recombinant
Non-recombinant
Multi-parent
Number of
descendant
samples
10
1,000
10,000
B
Recombination nodes
0 20 40 60
0 100 200 300 400 500 600 700 800
Time to the MRCA of sc2ts parents
C
Recombination nodes
0 25 50 75 100 125
0 1 2 3 4 5 6 7 8 9 10 12 14 16 18
Pango distance between sc2ts parents